

Abir Hirani

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Skills

Languages: Python, C, Bash, SQL, Java

ML/AI Frameworks: PyTorch, TensorFlow, Transformers, BERT, Hugging Face, Scikit-learn

Libraries & Tools: OpenCV, NumPy, Librosa, PySide6, PyQt6, Flask, Jupyter Notebook

Platforms & Systems: Git, Unix/Linux, CUDA, Raspberry Pi, Arduino

Design: Fusion 360, Figma, Photoshop, Microsoft 360, Canva

Education

Bachelor of Computer Science, Honours Computer Science Co-op

Aug 2024 – Apr 2028

University of Windsor • ON

- **Focus:** Machine Learning and Large Language Models
- **Awards:** International Student Scholarship of Excellence, Global Engagement Award

Experience

Artificial Intelligence Intern (AI), Glendor

Aug 2026 – Present

- Evaluated performance of open-source LLMs including Llama, Claude, and Mistral across text and image processing tasks
- Developed **fine-tuning workflows for pre-trained** language models using custom datasets optimized for mobile deployment
- Implemented **model compression and optimization techniques** to adapt LLMs for resource-constrained mobile environments

Resident Assistant (RA), University of Windsor

Aug 2024 – Apr 2025

- Provided leadership and acted as primary point of support for **32 residents** while maintaining **24/7 on-call availability** for emergencies and crisis situations
- Organized **8+ community-building events** per semester including educational workshops, social activities, and wellness programs to foster inclusive residence community
- Collaborated with residence staff and campus resources to ensure student wellbeing, **consistently meeting administrative deadlines and documentation requirements**

Projects

Text Classification with Transformers

Jan 2025 – Apr 2025

- Implemented **transformer-based BERT model** for **multi-class document categorization** using **Hugging Face Transformers**
- Fine-tuned **pretrained models** with **tokenization, attention masks, and learning rate scheduling** to optimize performance
- Developed **sarcasm detection model** on **28,000 news headlines** achieving **85.65% classification accuracy**

Lip Reading Machine Learning Model

May 2024 – Aug 2024

- Built **deep learning pipeline** using **CNN-LSTM architecture** to predict spoken words from **lip movements** in video sequences
- Preprocessed data with **OpenCV** for **face detection, lip region extraction**, and frame normalization to improve accuracy
- Optimized performance with **data augmentation, transfer learning, and hyperparameter tuning** for **visual speech recognition**

DriveCore – RC Vehicle Control System

Jan 2024 – Apr 2024

- Won "**Most Innovative Project**" award at University of Windsor Fall 2024 **Demo Day** among **50+ competing projects**
- Engineered a remote-control system with **real-time video streaming, sensor integration**, and custom **GUI** using **PySide6**
- Developed low-latency **UDP communication protocol** between **Raspberry Pi** and client PC for responsive control
- Integrated **OpenCV-based computer vision** for **obstacle detection, depth estimation**, and **Kalman filter** path smoothing

Volunteer Experience

Head of Projects, AI club, University of Windsor

- Led planning and execution of multiple community initiatives, coordinated team members, and ensured project milestones were delivered on time
- Managed project schedules, delegated tasks, and oversaw quality of deliverables to support organizational goals